

functionality, and `TimeLib` for handling clock features.

Once the libraries have been installed, the `TFT\_eSPI` library must be configured to support the GC9A01 display. This involves navigating to the Documents folder, opening the Arduino directory, accessing the Libraries folder, and locating the `TFT\_eSPI` folder within it. Inside it, the `User\_Setup.h` file must be opened, and the line corresponding to the GC9A01 display should be uncommented. The SPI pins must then be configured to match the setup on the IndusBoard.

Any available pins on the board may be used. In this case, pins 21, 1, 3, 4, 5, 6, 7, 8, and 9 were used

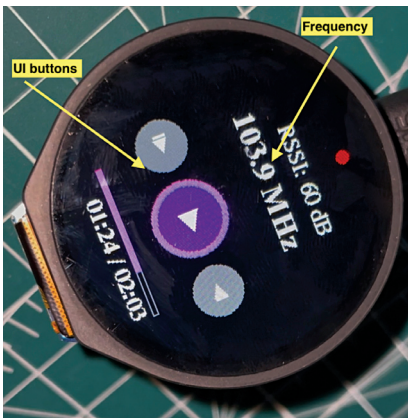


Fig. 4: FM radio UI

for the display and touch interface. These pin definitions should be correctly reflected in the code. Fig. 2 shows the configuration for the display in the `User\_Setup.h` file of the `TFT\_eSPI` library.

In the source code, the required libraries are included first, followed by the setup for the touchscreen driver. A basic user interface (UI) for the FM radio is then created, featuring three buttons: play/pause and two for changing channels. The code monitors touch coordinates to identify which button is pressed and performs the corresponding action, such as changing the FM frequency or toggling play and pause.

Circuit and working

Fig. 5 shows the circuit diagram of the miniature FM device. It is built around the GC9A01 (MOD1), IndusBoard Coin V2 (MOD2), CST816 touch screen, stereo radio module RDA5807M FM, two speakers (SP1 and SP2), a 3.7V Li-Po battery, and a few other components.

The touch display and FM radio both utilise the I²C protocol for communication, allowing them to be connected either to the same I²C port on the IndusBoard Coin or to separate ports. For simplicity, different ports

are used in this setup. The SDA and SCL pins for the touch display are defined in the code and connected accordingly, as shown in the circuit. The touch driver uses hardware I²C pins 8 and 9. The display itself uses SPI and is connected to the IndusBoard's SPI pins, as defined in the `UserSetup` file earlier.

The IndusBoard Coin V2 provides a 3.3V power supply via the VIN or 3V pin, which powers the GC9A01 display, CST816S touch controller, and RDA5807M FM radio module. A common ground (GND) is shared by all components to ensure proper operation.

Construction

For FM audio output, either an audio jack or speaker can be connected to the speaker audio-out pin. An audio jack is recommended, allowing either earphones or a speaker to be used as preferred. In this device, a speaker with an inbuilt amplifier has been used. Both speakers (SP1 and SP2) are connected to the LOUT and ROUT pins of the RDA5807M to provide stereo audio output.

To assemble this compact wearable FM device, the components are soldered onto the back side of the IndusBoard Coin V2, while the

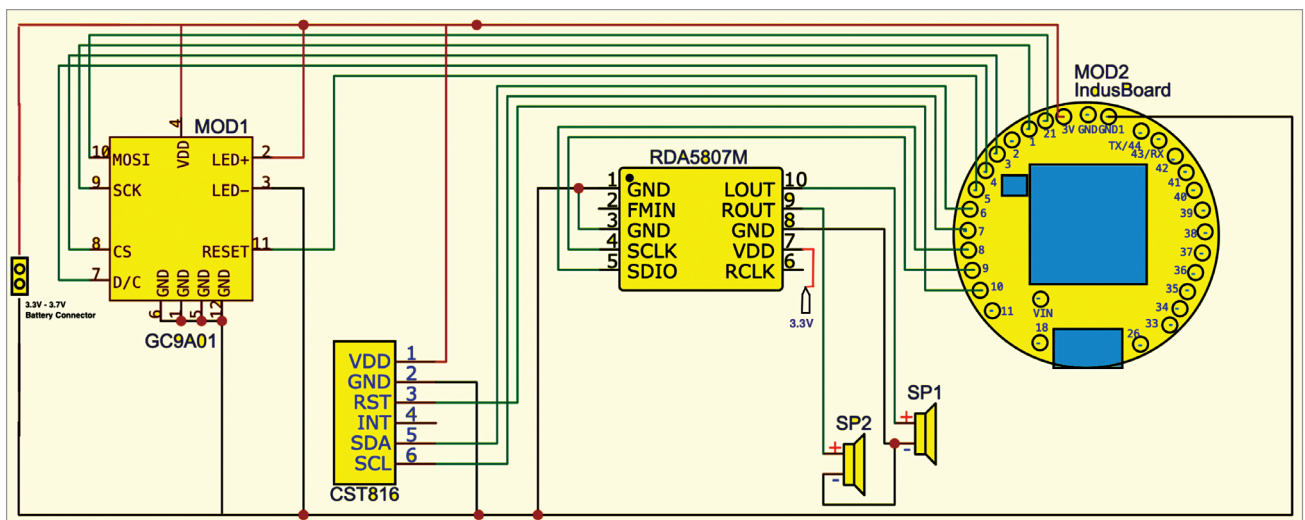


Fig. 5: Circuit diagram